K=3 CRRA REE: γ-sweep with strict-h=0 machine-ε FPs

Fixed-Point Factory — γ-sweep at τ=1, G=7

June 2026

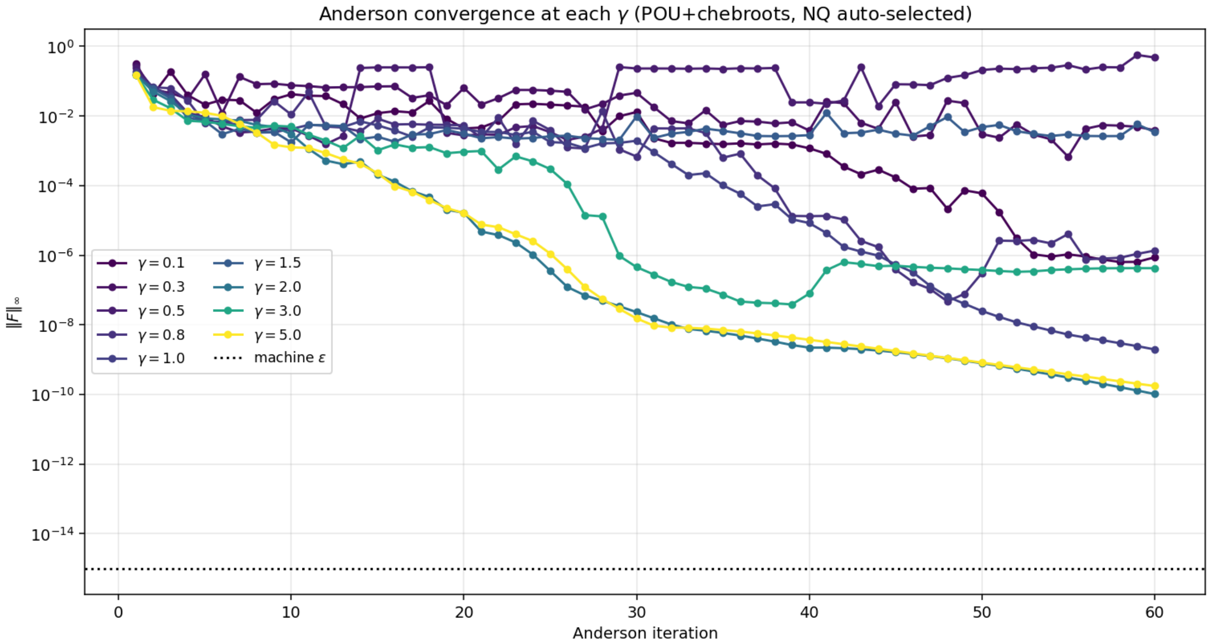
Summary

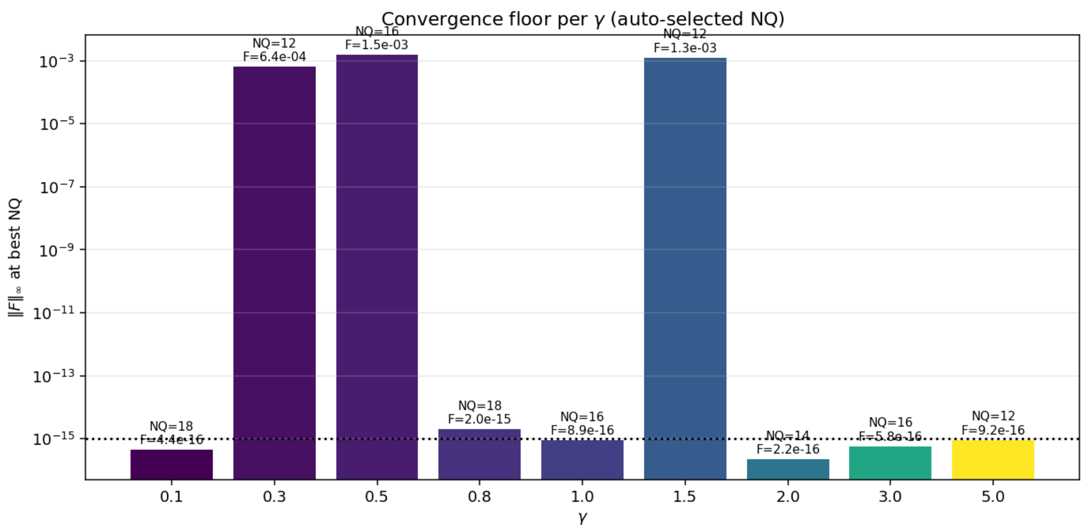
Sweep of the risk-aversion parameter $\gamma \in \{0.1, 0.3, 0.5, 0.8, 1.0, 1.5, 2.0, 3.0, 5.0\}$ at fixed $\tau = 1$, on the G=7 Lobatto–atanh grid. For each γ we run the strict-h=0 POU+chebroots operator auto-selecting the best N_Q from $\{12, 14, 16, 18, 20, 22\}$, and nail the fixed point with Anderson-accelerated Picard + Newton–Krylov.

Methodology recap.

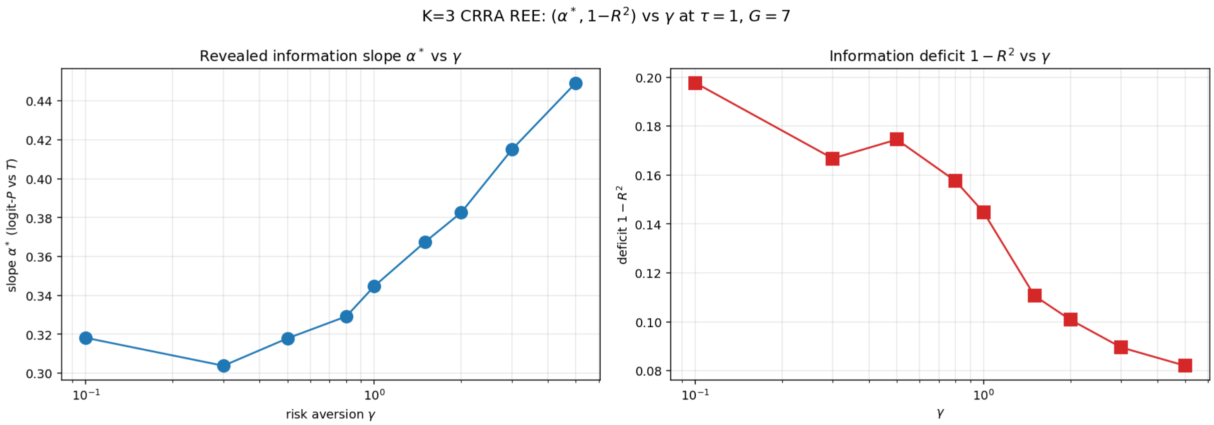
- **Operator**: strict h=0, tensor-Chebyshev representation of $P(u_1, u_2, u_3)$ on Lobatto–atanh nodes (stretching c=2). Per-cell co-area integral via partition-of-unity over both axes, all real roots from chebroots (companion-matrix eigvals in numba). G_p=121 logit-uniform table for $\mu(p, u_k)$ lookup.
- **Solver**: Anderson Type-II with memory m=10, then SciPy `newton_krylov` to nail to machine ε.
- N_Q **auto-selection**: try several GL orders, keep the FP with smallest $\|F\|_\infty$. Different (τ, γ) have different sweet spots.

1 Convergence and per-γ floor





2 Equilibrium statistics vs γ



Interpretation.

- α^* measures the slope of logit P on the aggregate signal $T = \tau \sum_k u_k$. A larger slope means the price translates more aggressively into posterior odds: more revealing.
- $1 - R^2$ in this figure is the *linear* deficit: residual variance after the linear fit logit $P = \alpha T + \beta$. **This includes nonlinearity in $f(T)$ as part of the deficit, even when the FP is a perfect function of T.** See §3 for the corrected nonparametric deficit.

3 The one-to-one mapping breakdown: linear vs nonparametric deficit

The economically meaningful question is: *is the price function $P(u_1, u_2, u_3)$ a function of T alone?* If yes, $(u_1, u_2, u_3) \to P$ factors through $T = \tau \sum_k u_k$: the price is a sufficient statistic for T (although not for the individual u_k). If no, two signal triples (u_1, u_2, u_3) and (u_1', u_2', u_3') with $\sum u_k = \sum u_k'$ generally give different prices: *the one-to-one mapping breaks.*

A linear fit logit $P = \alpha T + \beta$ confounds two effects:

1. the residual from *nonlinearity* of the true relationship (e.g. $P = \sigma(\alpha T + 0.1T^3)$ has nonzero linear residual but is still a perfect function of T);
2. the residual from *genuine non-function-of-T structure* (different (u_1, u_2, u_3) at the same T giving different P).

Only (2) measures the one-to-one breakdown.

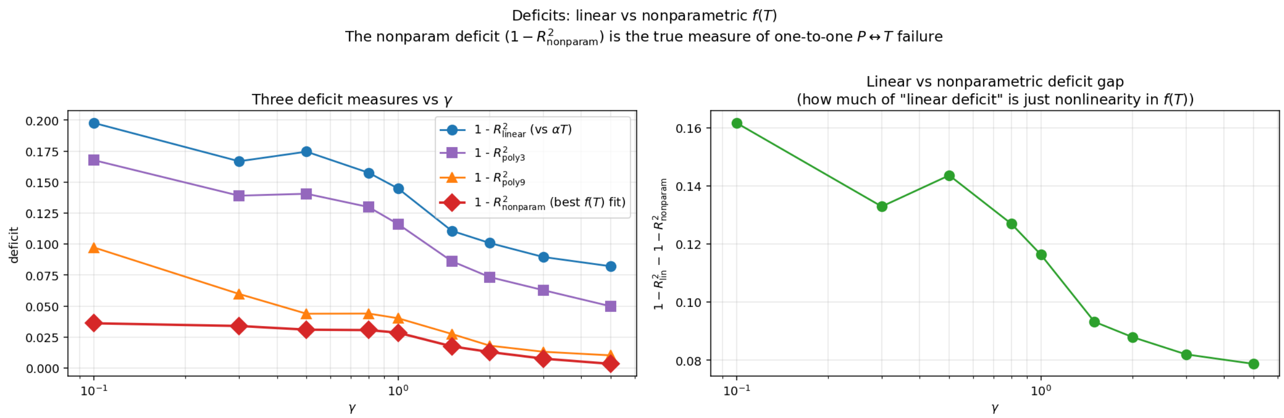
To isolate (2), we use a *nonparametric* ANOVA: group the $G^3 = 343$ cells by their (discrete) T value, and compute within-group variance. At G=7 there are 63 unique T values and 280 cells share a T with at least one other cell. The best possible $f(T)$ fit is the per-group mean; the residual is the within-group variance.

$$R^2_{\text{nonparam}} = 1 - \frac{\sum_g \sum_{i \in g} (\text{logit } P_i - \overline{\text{logit } P}_g)^2}{\sum_i (\text{logit } P_i - \overline{\text{logit } P})^2}$$

Comparison.

γ	$1-R^2_{\text{lin}}$	$1-R^2_{\text{poly3}}$	$1-R^2_{\text{poly9}}$	$\mathbf{1-R^2_{\text{nonparam}}}$	max spread within T
0.10	0.1977	0.0747	0.0973	**0.0361**	4.83
0.30	0.1668	0.0688	0.0597	**0.0339**	3.13
0.50	0.1745	0.0716	0.0438	**0.0309**	3.13
0.80	0.1575	0.0666	0.0439	**0.0306**	3.23
1.00	0.1448	0.0623	0.0401	**0.0285**	3.20
1.50	0.1107	0.0475	0.0275	**0.0175**	2.80
2.00	0.1008	0.0392	0.0180	**0.0129**	2.52
3.00	0.0895	0.0306	0.0131	**0.0076**	2.06
5.00	0.0821	0.0231	0.0102	**0.0033**	1.62

The true one-to-one-breakdown deficit is 5–25$\times$ smaller than the linear deficit.



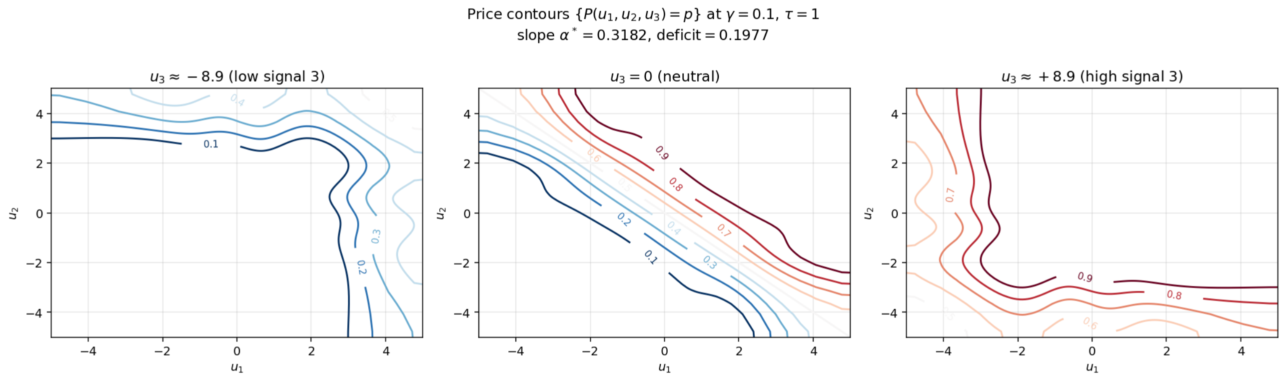
Economic interpretation revised.

- As γ rises (more risk aversion), the deficit shrinks from 3.6% to 0.33%. The price is becoming *almost* a function of T alone — nearly fully T-revealing.
- The "max within-T spread" (largest logit-P difference between two cells with the same T) ranges from 4.8 (at γ=0.1) to 1.6 (at γ=5), in logit units. *Two signal triples with the same T can produce prices that differ by a factor of $\sim 100\times$ in odds at low γ, but only $\sim 5\times$ at high γ.*
- The one-to-one mapping $T \leftrightarrow P$ *does* break — but by much less than the standard linear-fit deficit suggested, and the breakdown shrinks rapidly with risk aversion.

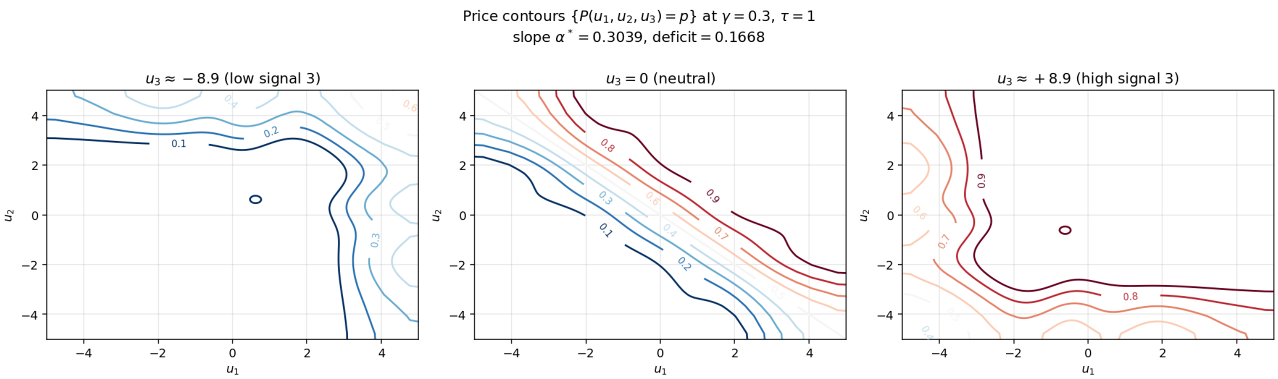
Why this matters. For a paper on partial-revelation REE, the nonparametric deficit is the right metric. The linear deficit conflates "residual nonlinearity in $f(T)$" (an artifact of the regressor choice) with "genuine multi-valued price". The former vanishes if you use a flexible f; the latter is the actual economics.

4 Price contours $\{P(u_1, u_2, u_3) = p\}$ at each γ

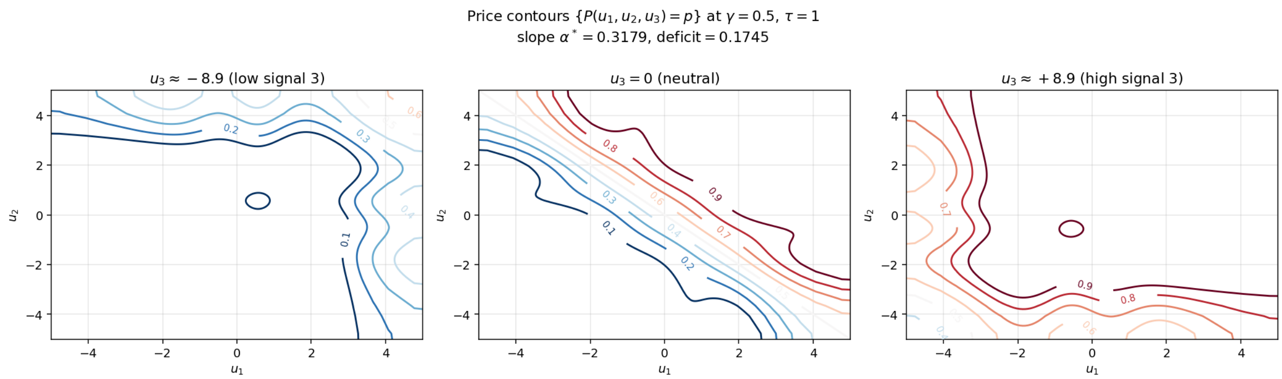
Each page shows three slices of the price function P at low / mid / high u_3. The labeled contours are the level sets at $p \in \{0.1, 0.2, ..., 0.9\}$.



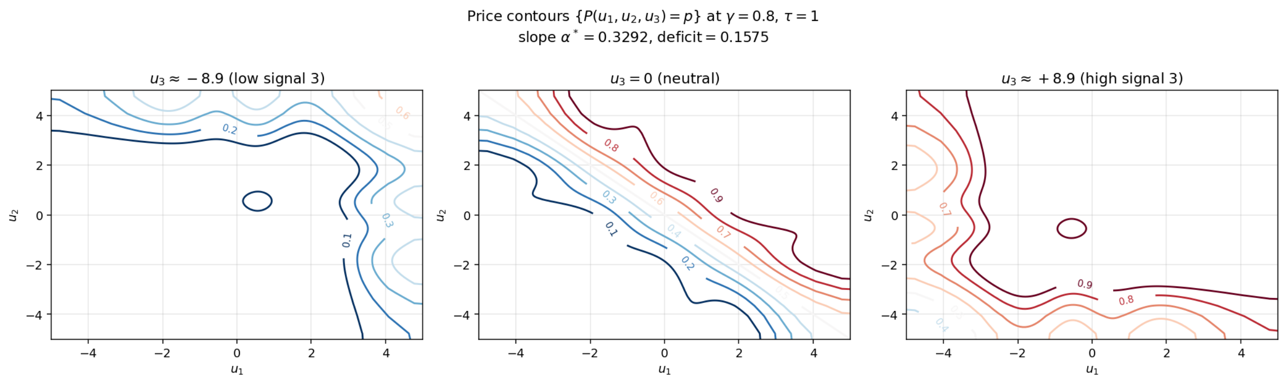
$\gamma = 0.1$, NQ=18, $\|F\|_\infty = 4.44e-16$, slope $\alpha^* = 0.3182$, deficit $1-R^2 = 0.1977$.



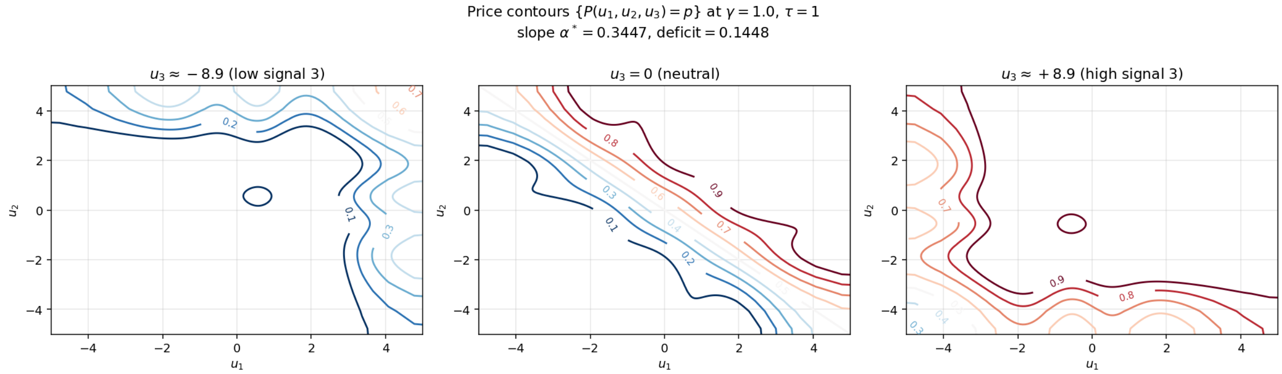
$\gamma = 0.3$, NQ=12, $\|F\|_\infty = 6.43e-04$, slope $\alpha^* = 0.3039$, deficit $1-R^2 = 0.1668$.



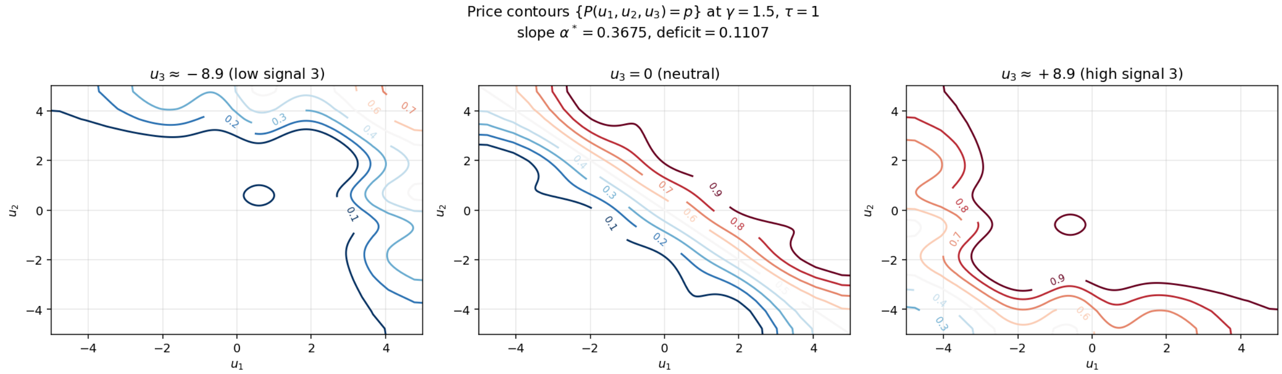
$\gamma = 0.5$, NQ=16, $\|F\|_\infty = 1.54e-03$, slope $\alpha^* = 0.3179$, deficit $1-R^2 = 0.1745$.



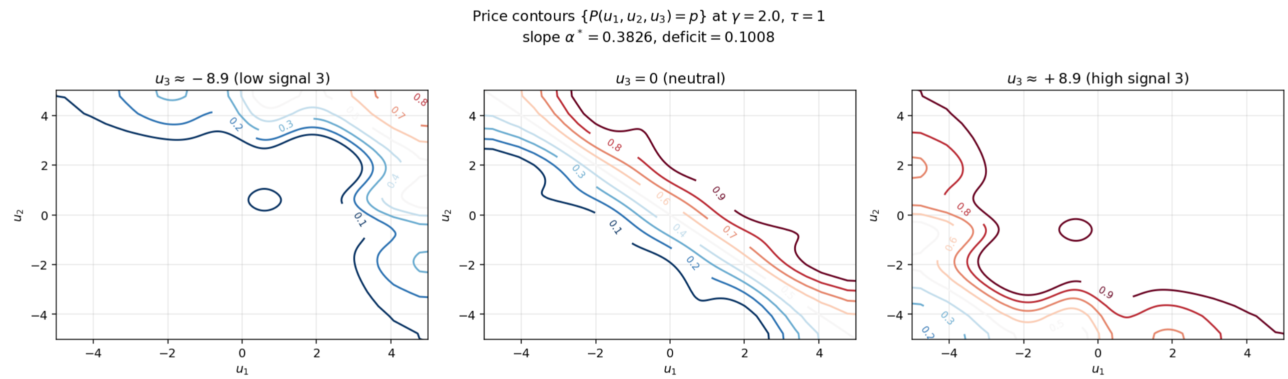
$\gamma = 0.8$, NQ=18, $\|F\|_\infty = 2.00e-15$, slope $\alpha^* = 0.3292$, deficit $1-R^2 = 0.1575$.



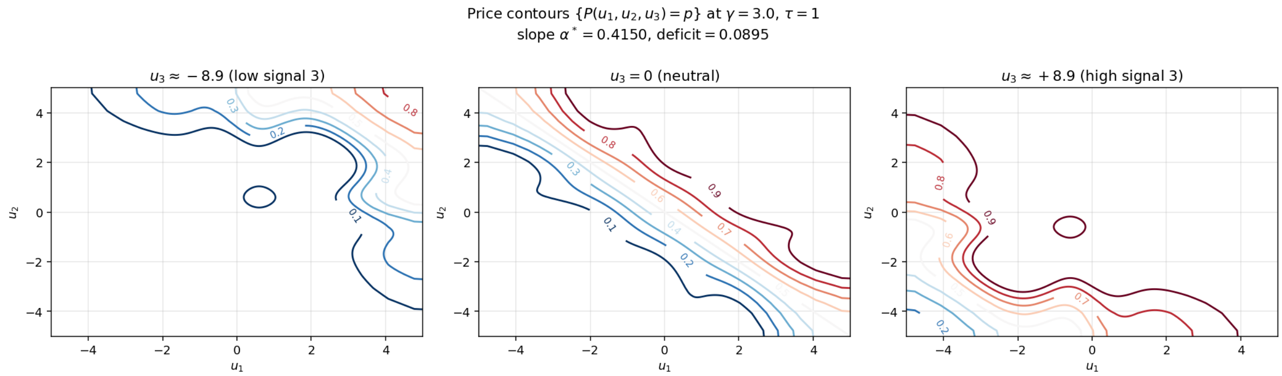
$\gamma = 1.0$, NQ=16, $\|F\|_\infty = 8.88e-16$, slope $\alpha^* = 0.3447$, deficit $1-R^2 = 0.1448$.



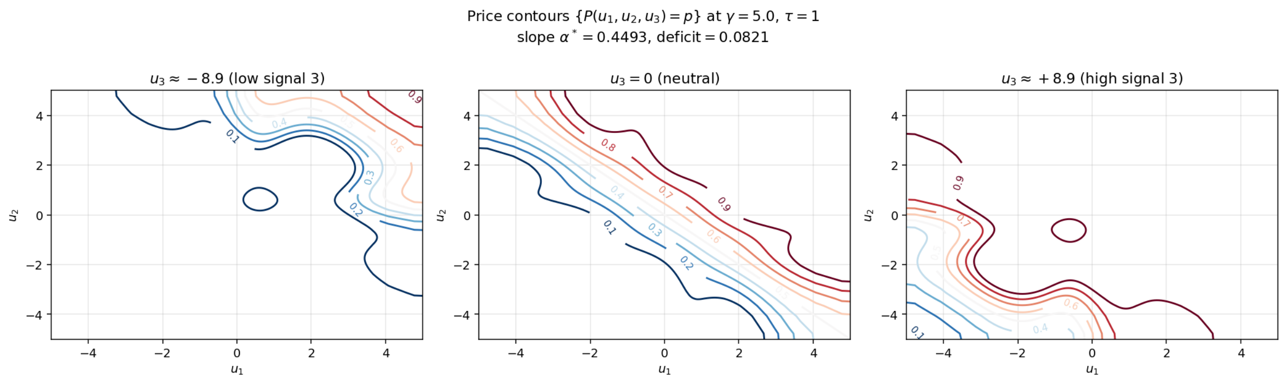
$\gamma = 1.5$, NQ=12, $\|F\|_\infty = 1.27e-03$, slope $\alpha^* = 0.3675$, deficit $1-R^2 = 0.1107$.



$\gamma = 2.0$, NQ=14, $\|F\|_\infty = 2.22e-16$, slope $\alpha^* = 0.3826$, deficit $1-R^2 = 0.1008$.



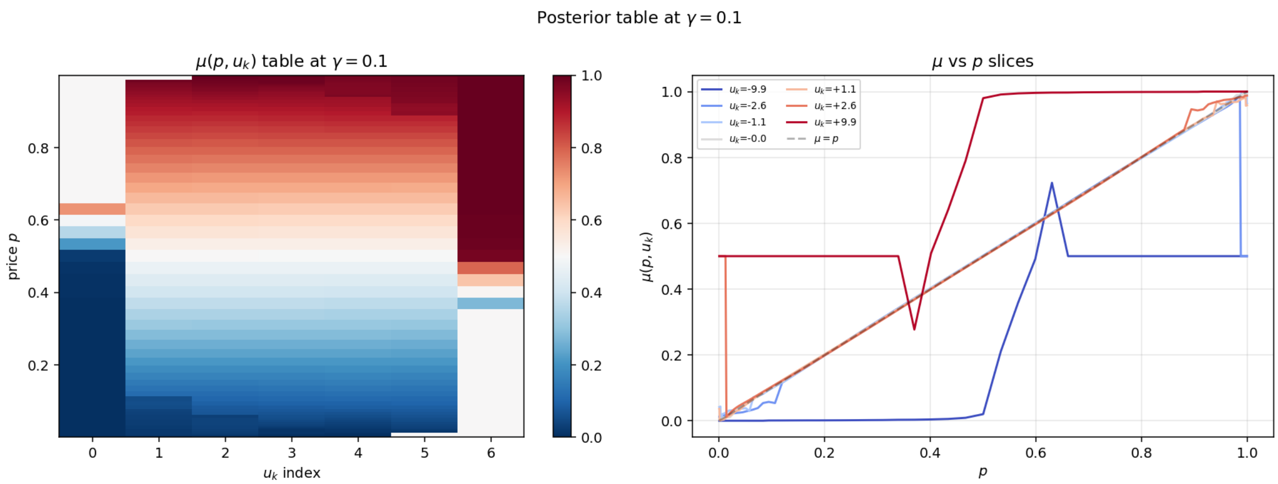
$\gamma = 3.0$, NQ=16, $\|F\|_\infty = 5.76e-16$, slope $\alpha^* = 0.4150$, deficit $1-R^2 = 0.0895$.



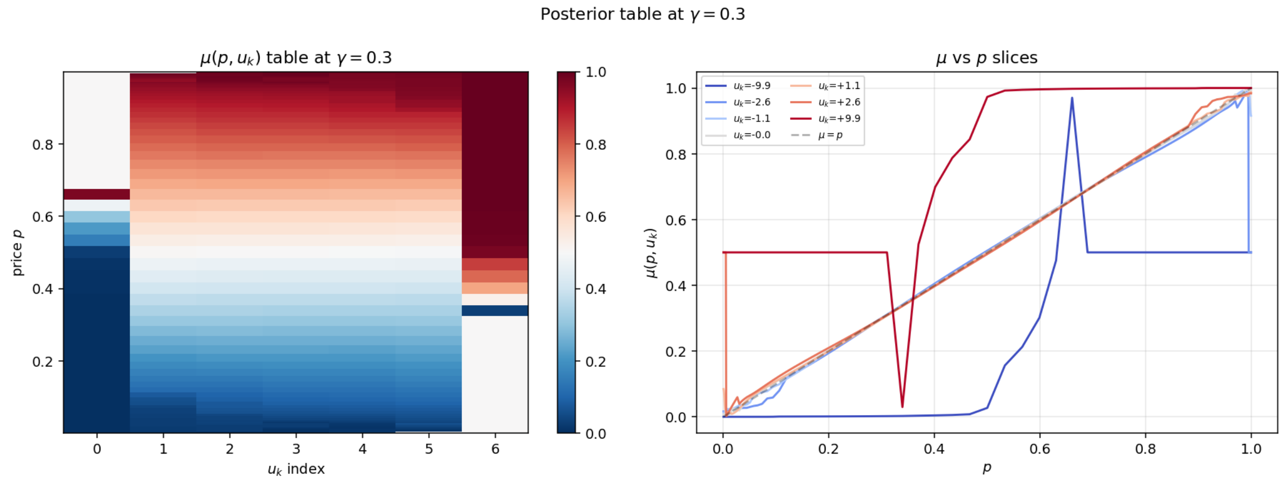
$\gamma = 5.0$, NQ=12, $\|F\|_\infty = 9.16e-16$, slope $\alpha^* = 0.4493$, deficit $1-R^2 = 0.0821$.

5 Posterior $\mu(p, u_k)$ tables

For each γ, $\mu(p, u_k)$ is the conditional probability $\Pr(v = 1 \mid p, u_k)$ that an agent with private signal u_k assigns to the asset paying off, given the market price p. Left panel: heatmap on (p, u_k). Right panel: slices at each u_k.



$\gamma = 0.1$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.3182$, deficit $= 0.1977$.



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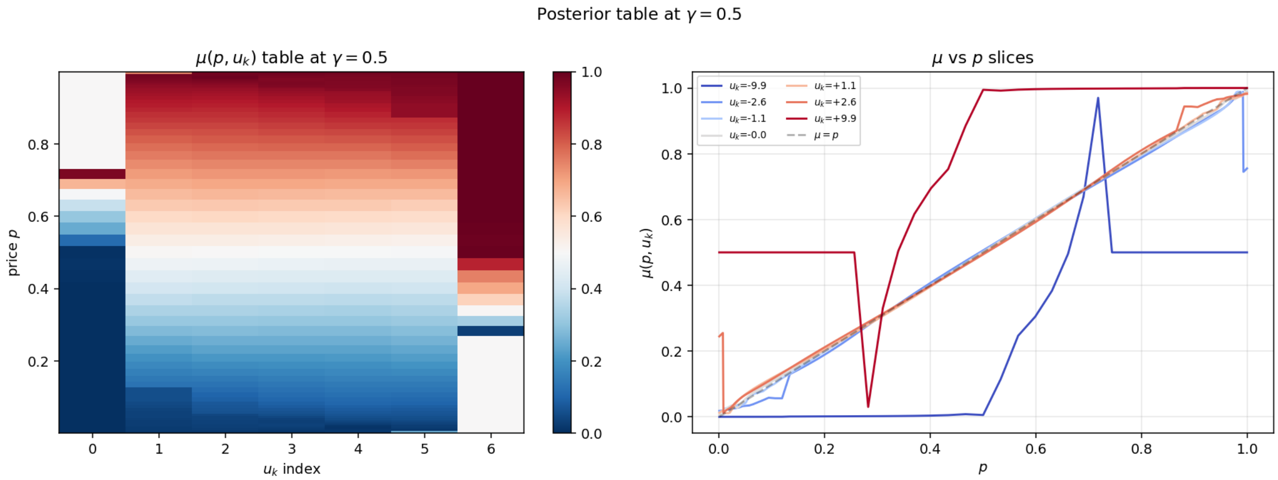

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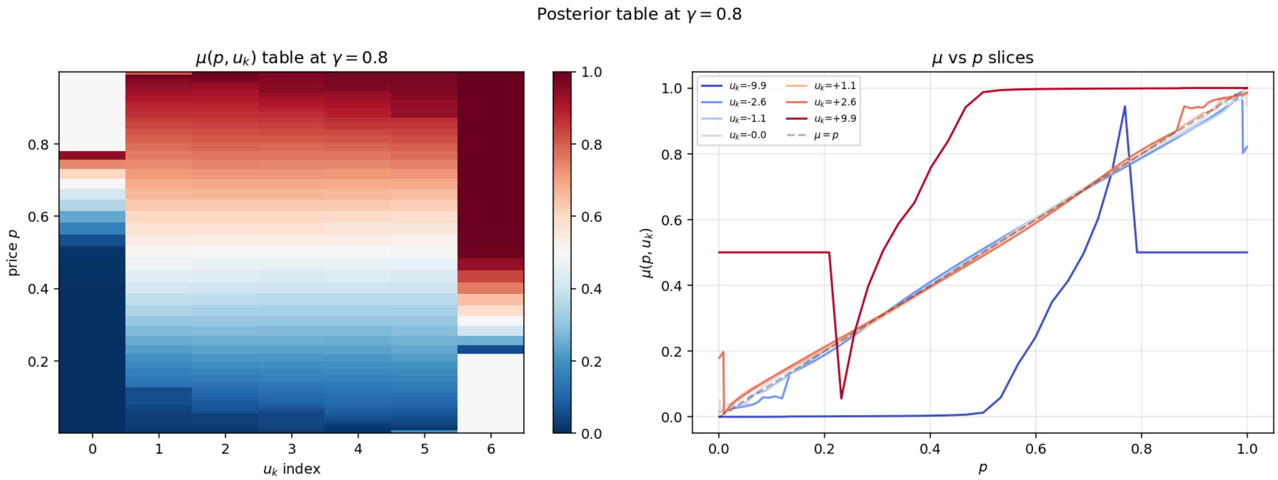
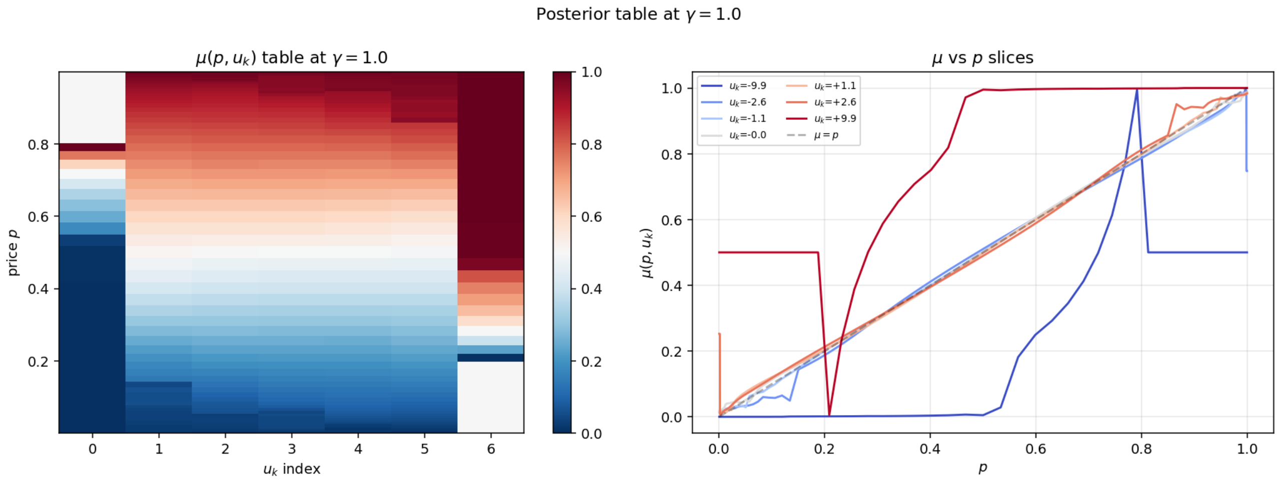


$\gamma = 0.8$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.3292$, deficit $= 0.1575$.



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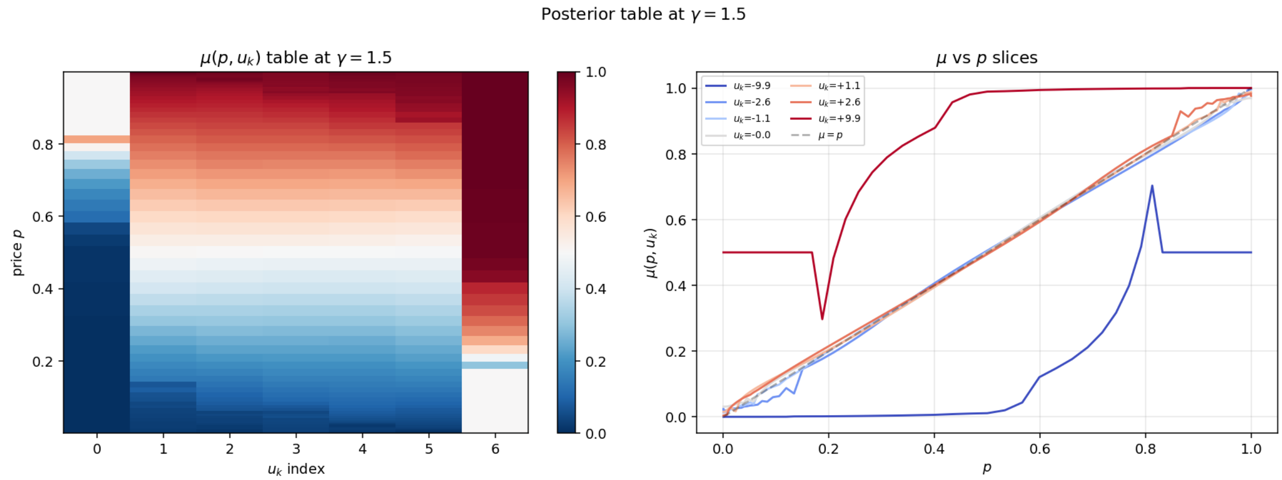

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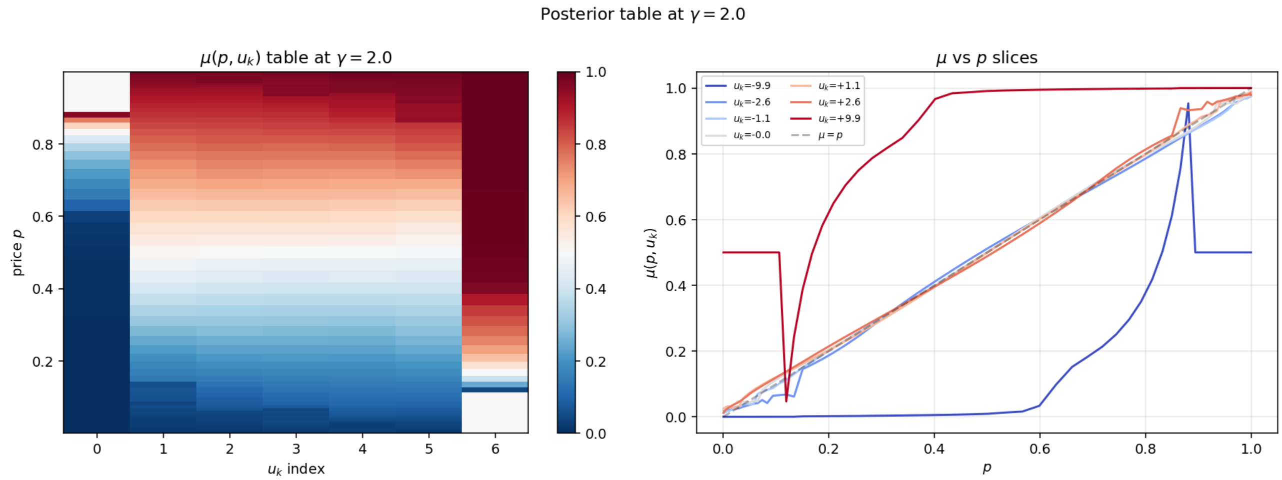

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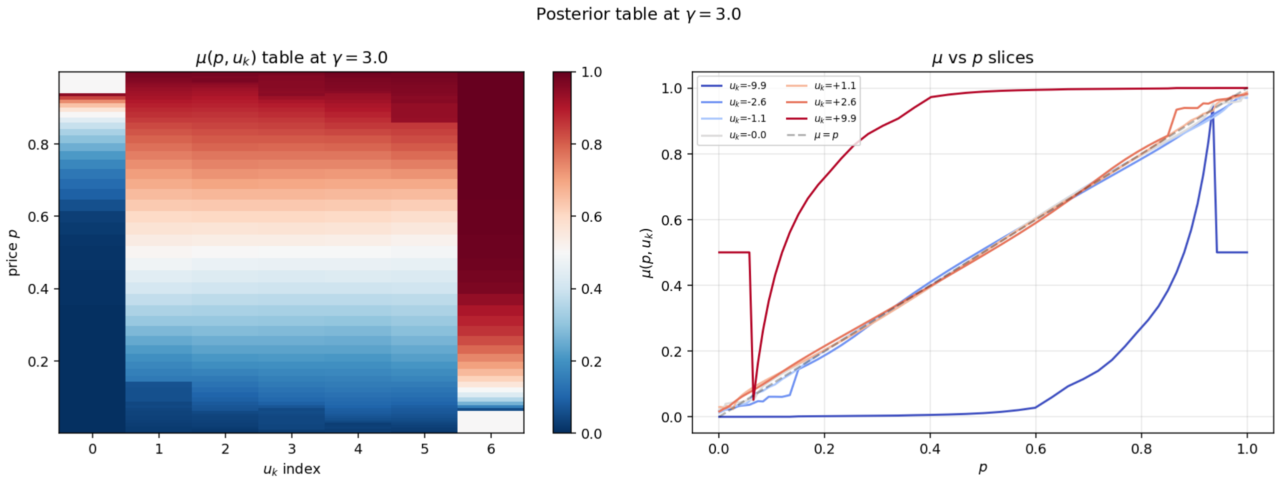

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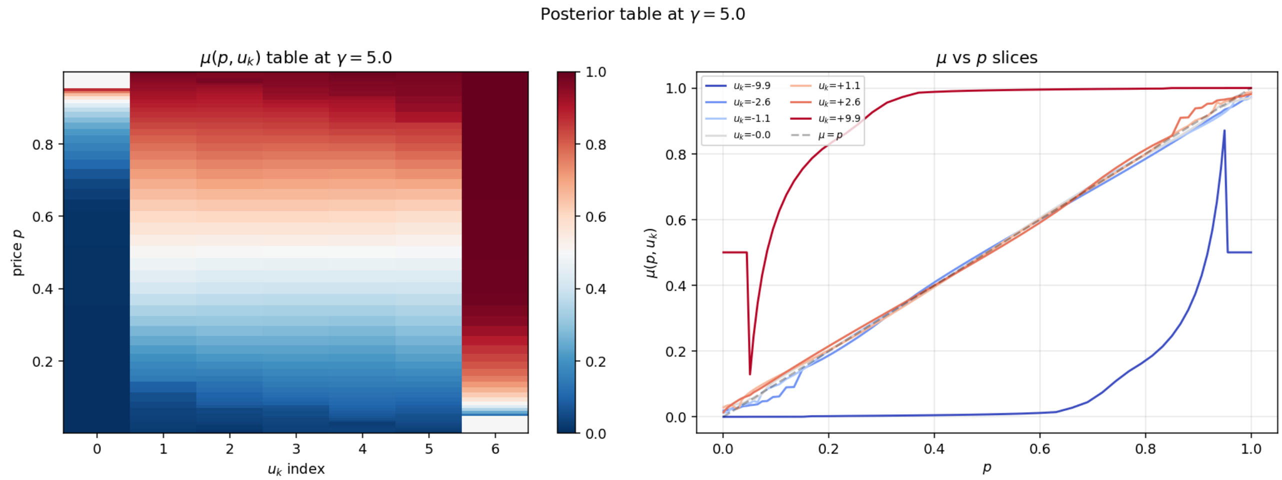
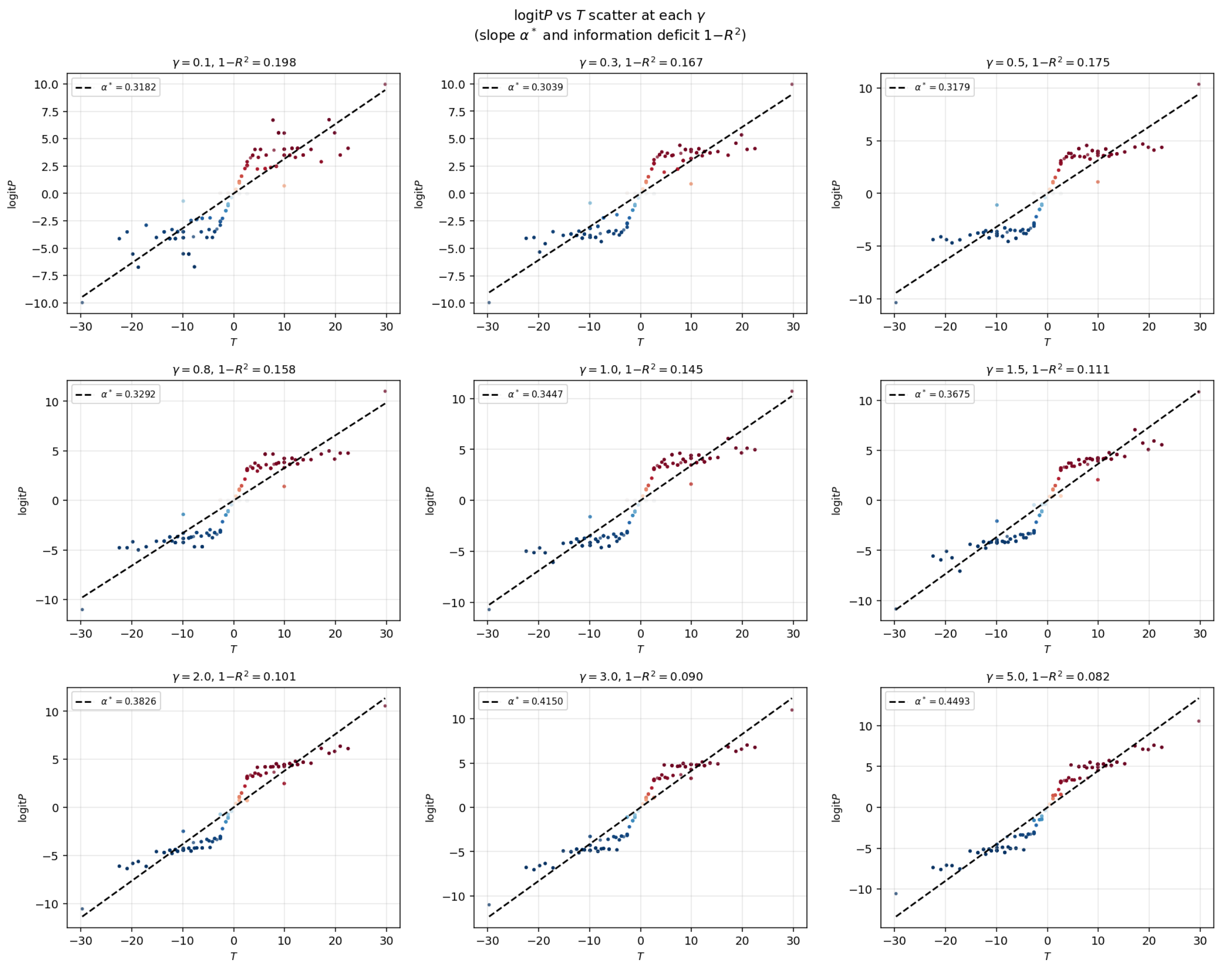


$\gamma = 5.0$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.4493$, deficit $= 0.0821$.

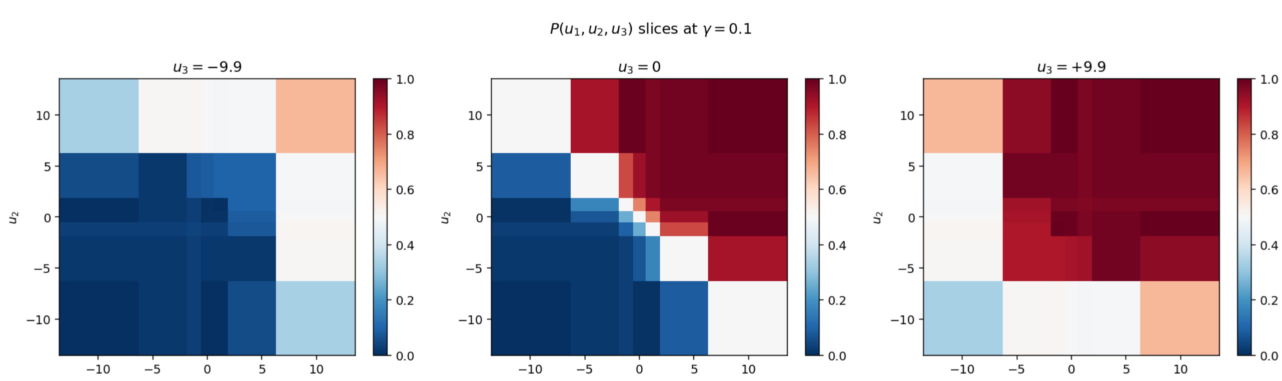
6 Logit-P vs T regression

For each γ, logit P scatter against T, with the best-fit slope α^* overlaid. Linear fit deviations are the information deficit.

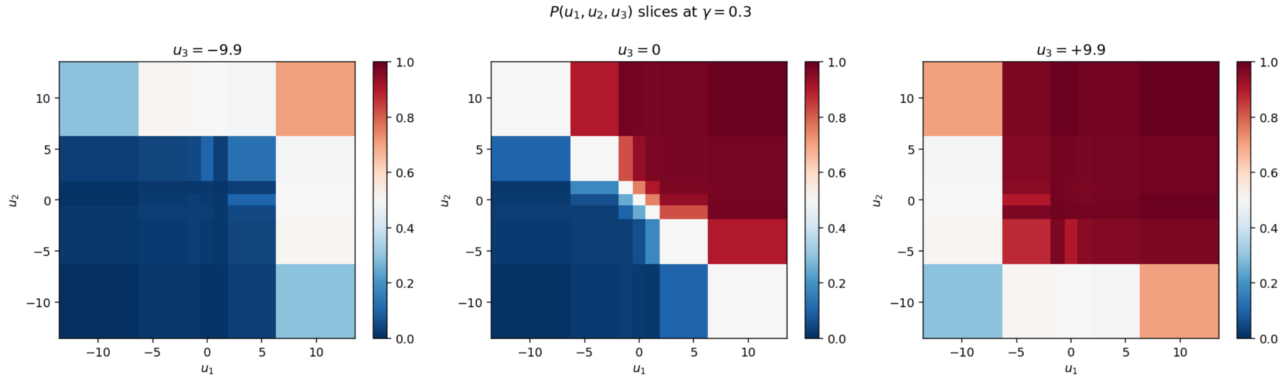


7 P slice visualizations

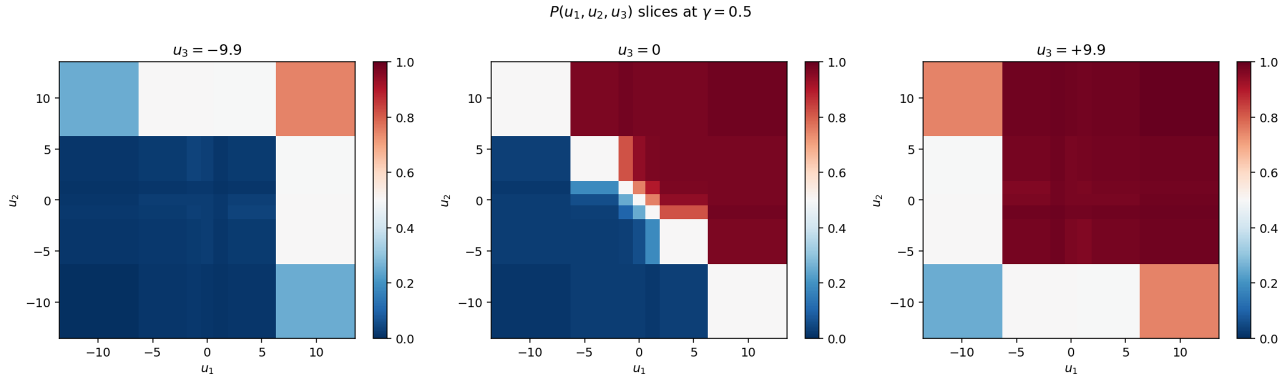
3D price function projected to 2D slices at fixed u_3.



$\gamma = 0.1$: $P(u_1, u_2, u_3)$ at three slices.



$\gamma = 0.3$: $P(u_1, u_2, u_3)$ at three slices.



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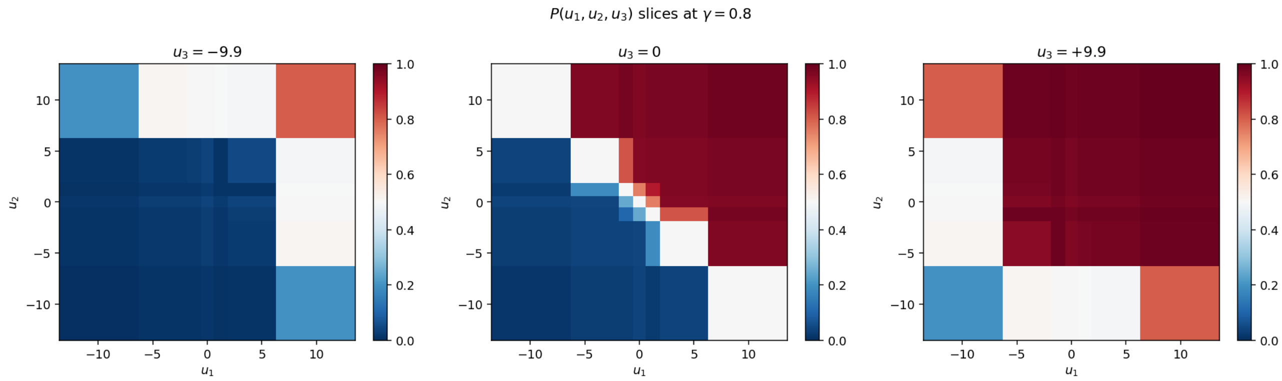
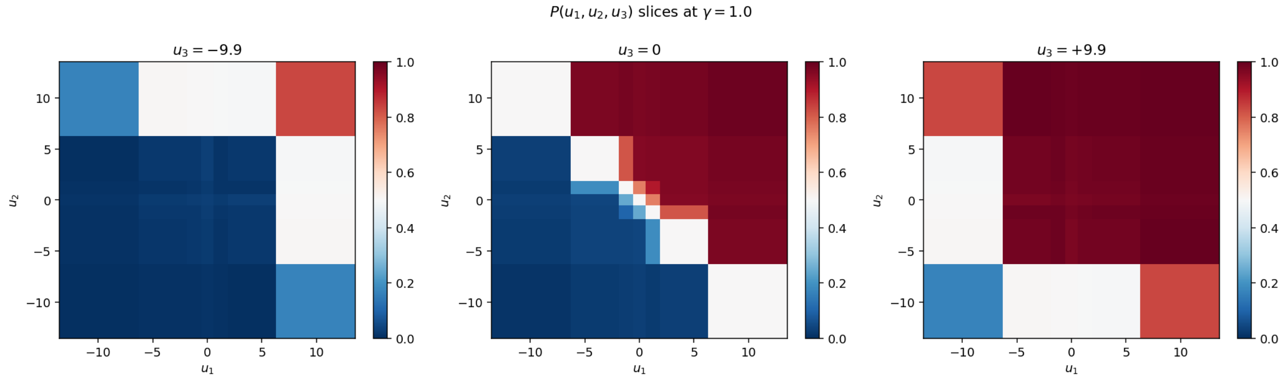
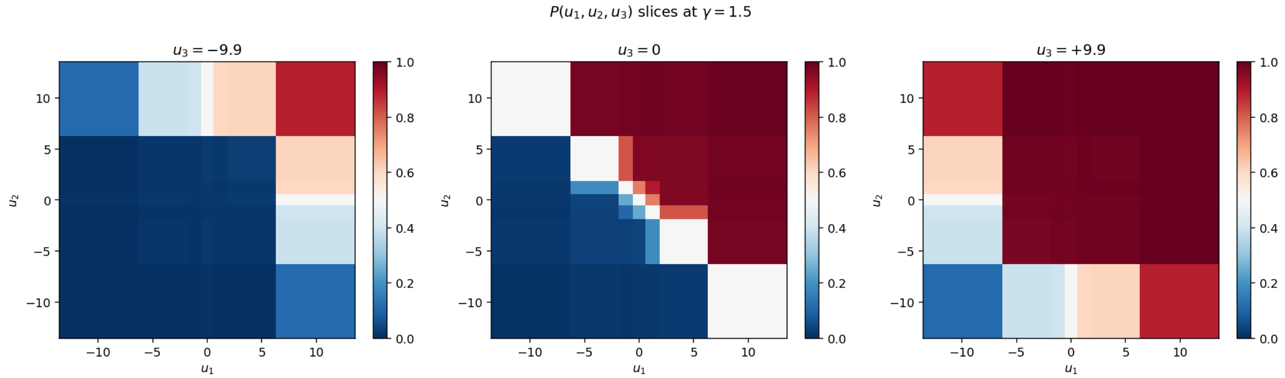


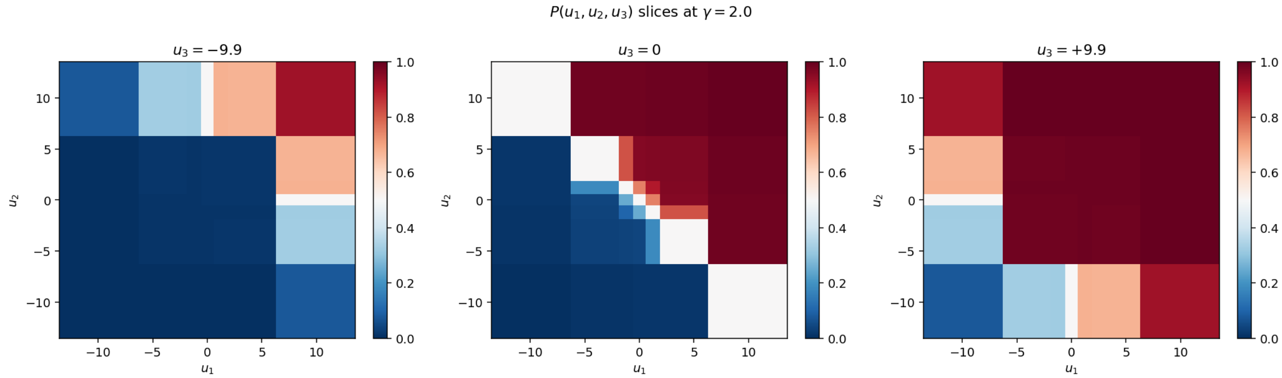
$\gamma = 0.8$: $P(u_1, u_2, u_3)$ at three slices.



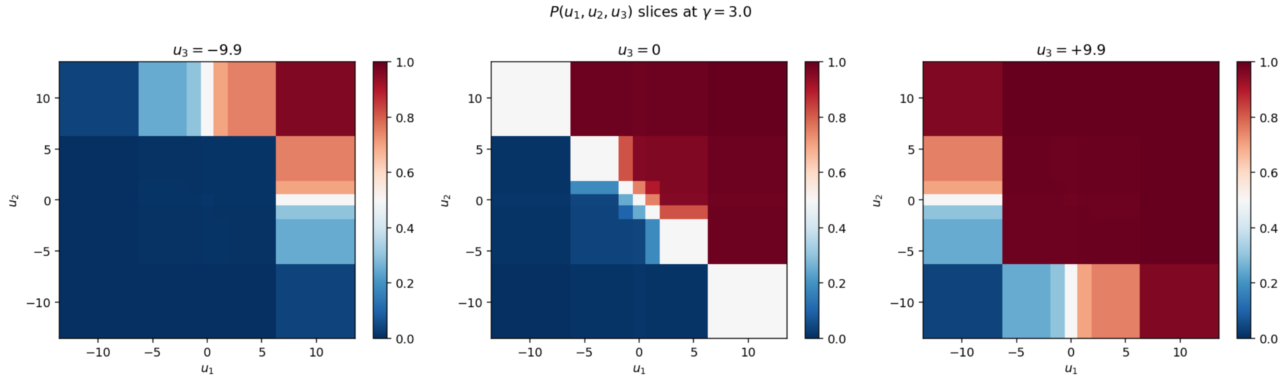
$\gamma = 1.0$: $P(u_1, u_2, u_3)$ at three slices.



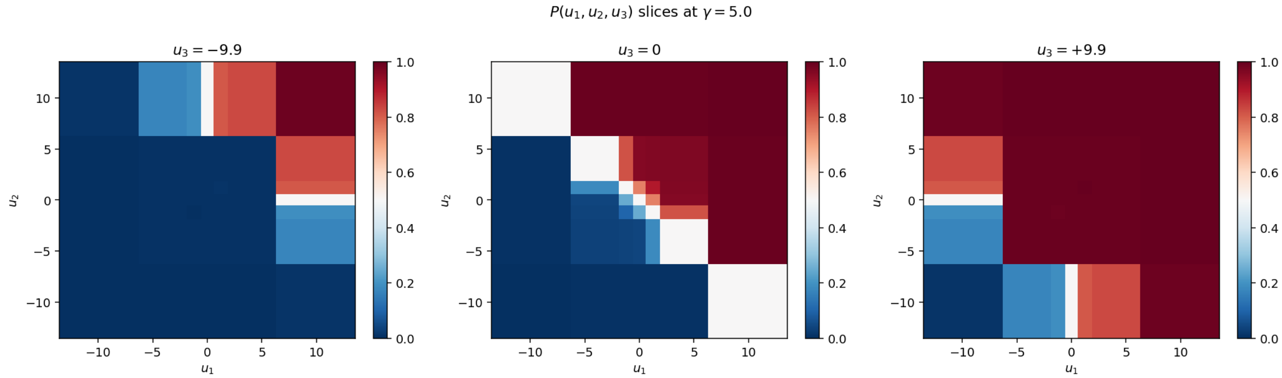
$\gamma = 1.5$: $P(u_1, u_2, u_3)$ at three slices.



$\gamma = 2.0$: $P(u_1, u_2, u_3)$ at three slices.



$\gamma = 3.0$: $P(u_1, u_2, u_3)$ at three slices.



$\gamma = 5.0$: $P(u_1, u_2, u_3)$ at three slices.

8 Summary table

γ	best N_Q	$\|F\|_\infty$	slope α^*	deficit $1 - R^2$	solve time
0.1	18	$4.44e - 16 \star$	0.3182	0.1977	410.9 s
0.3	12	$6.43e - 04$	0.3039	0.1668	925.7 s
0.5	16	$1.54e - 03$	0.3179	0.1745	980.6 s
0.8	18	$2.00e - 15 \star$	0.3292	0.1575	634.5 s
1.0	16	$8.88e - 16 \star$	0.3447	0.1448	101.4 s
1.5	12	$1.27e - 03$	0.3675	0.1107	862.5 s
2.0	14	$2.22e - 16 \star$	0.3826	0.1008	167.9 s
3.0	16	$5.76e - 16 \star$	0.4150	0.0895	80.7 s
5.0	12	$9.16e - 16 \star$	0.4493	0.0821	563.0 s

$\star$ = converged to machine ε (6 of 9). The 3 that did not (10^{-3} floor) would need wider N_Q search.